



First Semester 2026-27

Course Handout Part II

Date: 31st July 2026

In addition to Part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number: AN F314
Course Title: Introduction to Flight
Instructor-in-Charge: Biswadip Shome (biswadip.shome@pilani.bits-pilani.ac.in)
Team of Instructors: N/A

1. Course description:

This course introduces the fundamental principles governing aircraft flight, including basic aerodynamics, atmospheric properties, aircraft performance, stability and control, propulsion systems, and Hypersonic vehicles. Students explore the forces acting on an aircraft and the engineering concepts that enable safe and efficient flight. The course provides a foundational understanding of flight mechanics and aircraft systems, preparing students for advanced studies in aerospace engineering.

2. Scope and Objective:

Following are the scope and objective of this course:

- Understand the fundamentals of flight: Introduce the basic principles of aerodynamics, the four forces of flight, and the physical concepts governing aircraft motion.
- Study aircraft systems and performance: Develop an understanding of aircraft structures, propulsion systems, stability, control, and performance under various operating conditions.
- Apply engineering principles: Use fundamental concepts of mathematics, physics, and engineering to analyse and solve introductory flight-related problems.
- Build a foundation for advanced aerospace studies: Prepare students for higher-level courses in flight mechanics, aircraft design, aerospace propulsion, and aviation engineering.

3. Course Learning Outcomes:

Upon successful completion of this course, the students will have the following learning outcomes:

- Explain the principles of flight by describing the four fundamental forces (lift, weight, thrust, and drag) and their interactions during flight.
- Analyse atmospheric effects on flight by interpreting how atmospheric properties influence aircraft performance and operations.
- Apply basic aerodynamic concepts to evaluate the performance of airfoil, wings, and aircraft under various flight conditions.



- d. Evaluate aircraft performance by estimating take-off, climb, cruise, descent, landing, range, and endurance characteristics using fundamental engineering principles.
- e. Describe aircraft stability and control and explain how control surfaces affect aircraft motion and manoeuvrability.
- f. Compare aircraft propulsion systems, including propeller- and jet-powered engines, and explain their operating principles and performance characteristics.
- g. Learn Hypersonic flight, concepts of Newtonian theory and its application to Hypersonic vehicles.
- h. Solve introductory flight-related engineering problems using mathematics, physics, and computational tools where appropriate.
- i. Interpret the relationship between aircraft design and flight performance, recognizing how design choices affect efficiency, safety, and operational capability.
- j. Communicate technical concepts related to flight through written reports, calculations, and engineering discussions

4. Textbooks and Reference Books:

TB: J.D. Anderson Jr., Introduction to Flight, 6th Edition, McGraw Hill, 2008.

RB: S.K. Ojha, Flight Performance of Aircraft, AIAA Education Series, 1995

5. Method of Conduct of the Course:

The course will be conducted using a combination of instructor-centred and learner-centred methods. The former would involve instructor-led class lectures and problem solving exercises while the latter would involve student-led collaborative group projects using simulation for experiential learning, technical report writing, and presentations.

6. Course Plan:

Week	Module	Lecture	Reference	Learning Outcome
1	M1: Introduction	1-3	Ch. 1-2 TB Class slides	• Understanding of historical developments in aviation • Understanding of aviation milestones • Understanding the components of an aircraft and types of aerial vehicles
2	M2: Standard Atmosphere	4-6	Ch. 3 TB Class slides	• Understanding of properties of standard atmosphere • Analyse how atmospheric conditions influence aircraft



Week	Module	Lecture	Reference	Learning Outcome
				performance and aerodynamic behaviour
3-4	M3: Basic Aerodynamics	7-12	Ch. 4 TB Class slides	• Understanding of fluid dynamic equations • Understanding the concept of Ideal fluid • Understanding viscous flows and the concepts of compressibility • Introductory understanding of wind tunnels and their operations
5-7	M4: Airfoil and Wings	13-19	Ch. 5 TB Class slides	• Understanding flow past bodies • Understanding flow separation • Understanding concepts of lift generation, drag, and moments • Understanding non-dimensional coefficients of lift, drag, and moment • Understanding airfoil and wings, concept of airfoil families • Understanding concepts of supersonic flight
7-11	M5: Aircraft Performance	20-31	Ch. 6 TB Class slides	• Understanding of aircraft performance • Comprehension of topics in aircraft performance: ▪ Aircraft drag polar ▪ Thrust and power ▪ Rate of climb ▪ Range and Endurance ▪ Altitude effects on aircraft performance ▪ Aircraft take-off and landing ▪ Aircraft turning flight
11-12	M6: Aircraft Stability and Control	32-35	Ch. 7 TB Class slides	• Understanding the concept of aircraft stability and control • Comprehension of following topics: ▪ Static stability ▪ Pitching moment ▪ Neutral point and static margin ▪ Directional and lateral static stability



Week	Module	Lecture	Reference	Learning Outcome
12-13	M7: Propulsion Systems	36-38	Ch. 9 TB Class slides	- Understanding of operating principles, performance, and application of aircraft propulsion systems, namely - Propeller-piston engine - Jet engines – Turbojet and Turbofan - Rocket engines
13-14	M8: Hypersonic Vehicles	39-40	Ch. 11 TB Class Slides	- Understanding of aerodynamic phenomena unique to hypersonic flow - Understanding of physical aspects of Hypersonic flow - Comprehension of Newtonian theory and its application to hypersonic flights

7. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Comments
Quiz	20 minutes	15%	TBA	- Best 04 out of 05 - Open textbook¹ (Publisher copy only) - Open class notes
Course project	-	15%	Report due date 15/11/2026	- Group project - Open resources
Mid-Semester exam	90 minutes	30%	09/10/26 (AN1)	- Open textbook¹ - Open class notes
Comprehensive exam	180 minutes	40%	14/12/26 (FN)	- Open textbook¹ - Open class notes

Notes:

- Only prescribed text book (Introduction to Flight by J.D. Anderson) and Hand-written class notes are allowed.
- Printed material, other books, photocopied materials, loose sheets are strictly prohibited. Use of any unapproved materials during the examination will be considered as malpractice and will invite strict disciplinary action as per the institute rules.



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3. Course project shall be taken in a group of maximum 3-5 members. The list of project topics will be shared by the IC. The evaluation of course project will be based on the report submitted and a presentation may be required. Project report submitted after due date will not be graded.

8. Chamber Consultation Hours: FD-II Room 2228, Wednesdays 3:00 PM to 4:00 PM.

9. Make-up Policy: Make up for Mid-semester and Comprehensive Examination will be conducted based on the requests received and as per the guidelines by AUGS/AGSR Division. No make-ups will be conducted for Quizzes.

10. Attendance Policy: As per guidelines by AUGS/AGSR Division.

11. Notices: Notices pertaining to this course will be displayed on Nalanda course page or be sent through email.

Prof. Biswadip Shome
Instructor-in-Charge AN F314